



	EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics Fall 2025			
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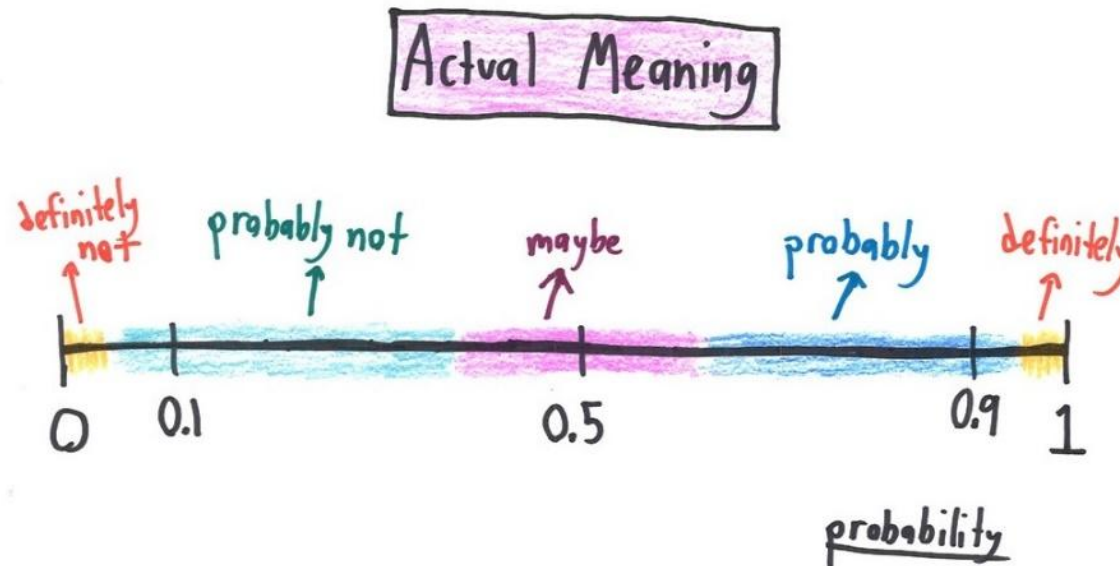


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 3 - Lecture No 04B – 1st September 2025



Announcements!

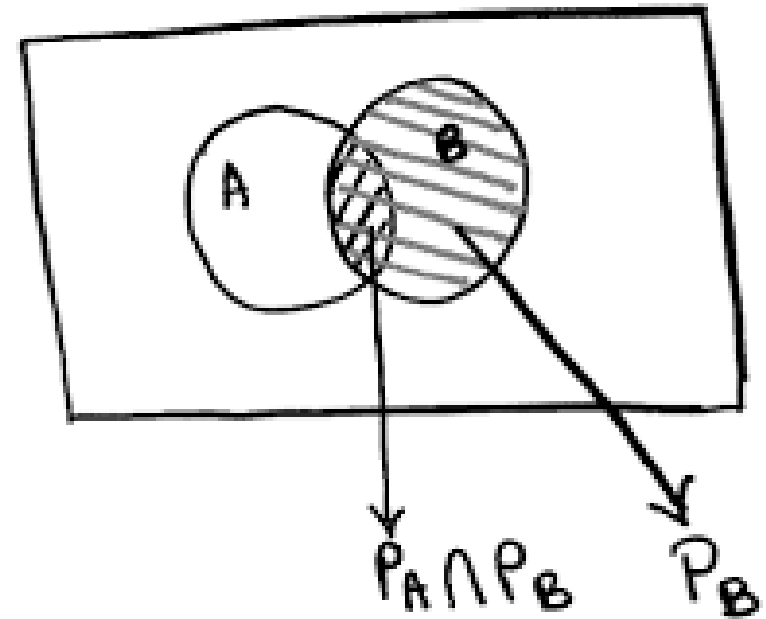
1. Lectures up to No 03 posted
2. Quiz No 01 – next class on September 03, 2025 (Wednesday) from Unit No 01

Agenda for today

1. Unit 2 - Conditional Probability and Bayes' Rule

Unit 2: Conditional Probability and Bayes' Rule

- Conditional Probability
- Three important Tools
 - Multiplication rule
 - Total probability theorem
 - Bayes' rule ($\rightarrow$ Inference)



Conditioning & Bayes' rule

- Let's say that you roll two dice and their sum comes out to be S .
 - ❖ If $S = 9$, what can we say about the first and second roll?
 - ✓ What is the probability that first roll is 1 or 2?

New knowledge should be used to modify our original beliefs.



Revised probabilities are called the conditional probabilities.



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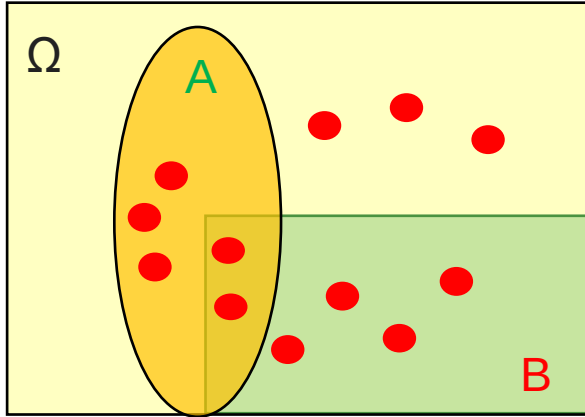


Revised probabilities are called the conditional probabilities.

The idea of conditioning

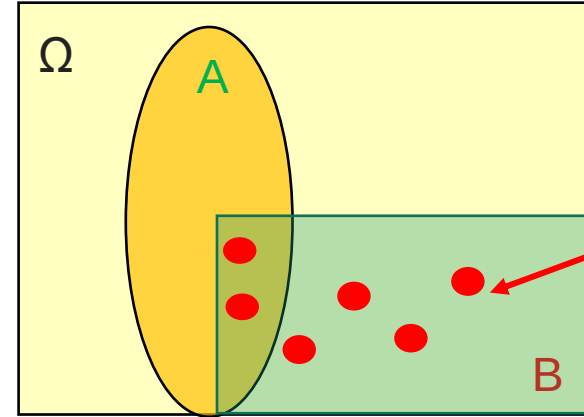
Use new information to revise a model

Assume 12 equally likely outcomes, each with probability $1/12$



$$P(A) = \frac{5}{12} \quad P(B) = \frac{6}{12}$$

If we are told that B occurred



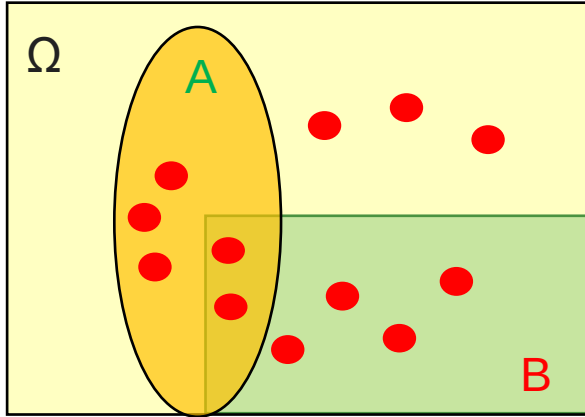
$$\frac{1}{6}$$

$$P(A|B) = \frac{2}{6} = \frac{1}{3} \quad P(B|B) = \frac{6}{6} = 1$$

The idea of conditioning

Use new information to revise a model

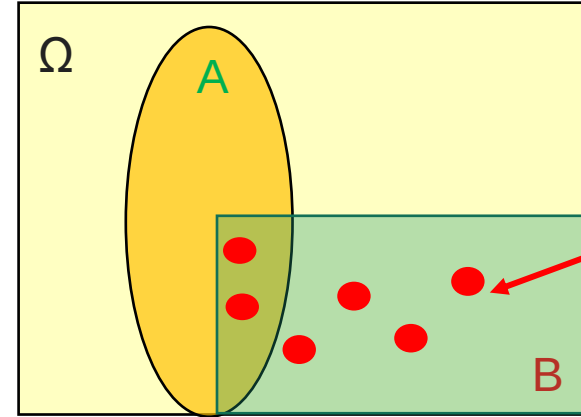
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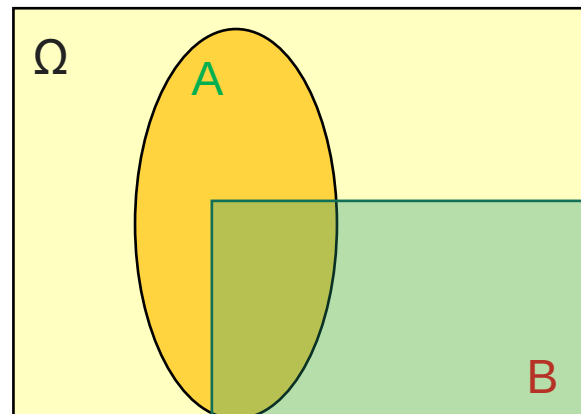
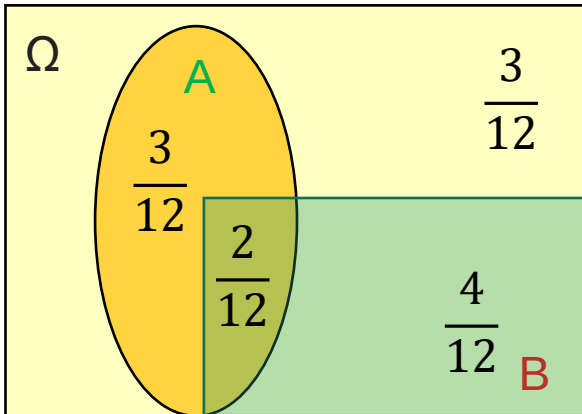
If we are told that B occurred



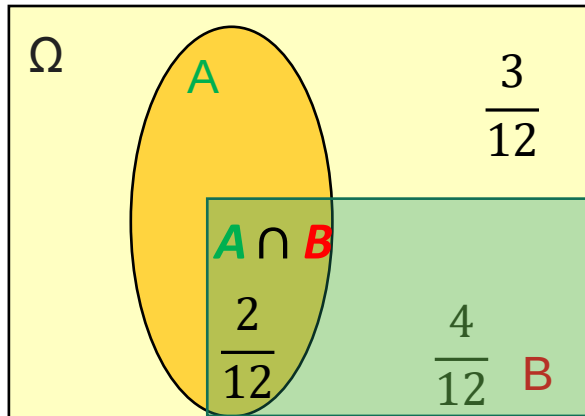
$$\frac{1}{6}$$

$$P(A|B) = \frac{2}{6} = \frac{1}{3} \quad P(B|B) = \frac{6}{6} = 1$$

Assume now that the events are not equally likely



Definition of conditional probability



- $P(A|B)$ = “probability of A , given that B occurred”

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/12}{4/12} = \frac{1}{2}$$

defined only when $P(B) > 0$

Example: two rolls of a 4-sided die

Y =
Second
roll

4				
3				
2				
1				
	1	2	3	4

X = First roll

- Let every possible outcome have probability $1/16$

- Let B be the event: $\min(X, Y) = 2$

Let $M = \max(X, Y)$

$$\rightarrow P(M = 1 | B) = 0$$

$$\rightarrow P(M = 3 | B) = \frac{P(M = 3 \text{ and } B)}{P(B)}$$

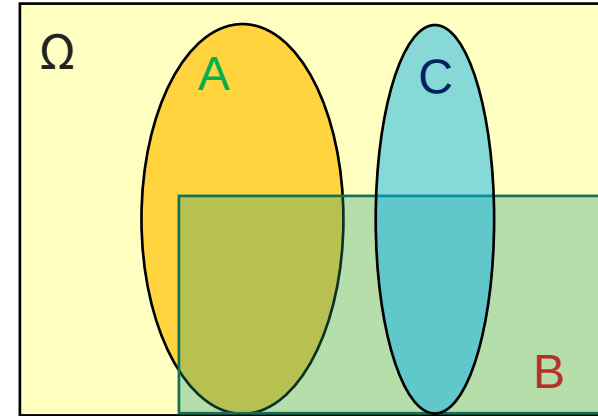
$$\rightarrow = \frac{2/16}{5/16} = \frac{2}{5}$$

Conditional Probabilities have the same properties as ordinary probabilities

→ • $P(A | B) \geq 0$ assuming $P(B) \geq 0$

→ • $P(\Omega | B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$

→ • $P(B | B) = \frac{P(B \cap B)}{P(B)} = 1$



• If $A \cap C = \emptyset$, then $P(A \cup C | B) = P(A | B) + P(C | B)$

→
$$= \frac{P((A \cup C) \cap B)}{P(B)} = \frac{P((A \cap B) \cup (C \cap B))}{P(B)} = \frac{P(A \cap B) + P(C \cap B)}{P(B)}$$

→
$$= P(A|B) + P(C|B)$$

→ Also finite countable additivity

Example: Models based on conditional probabilities.

- Event **A**: Airplane is flying above.
- Event **B**: Something registers on radar screen

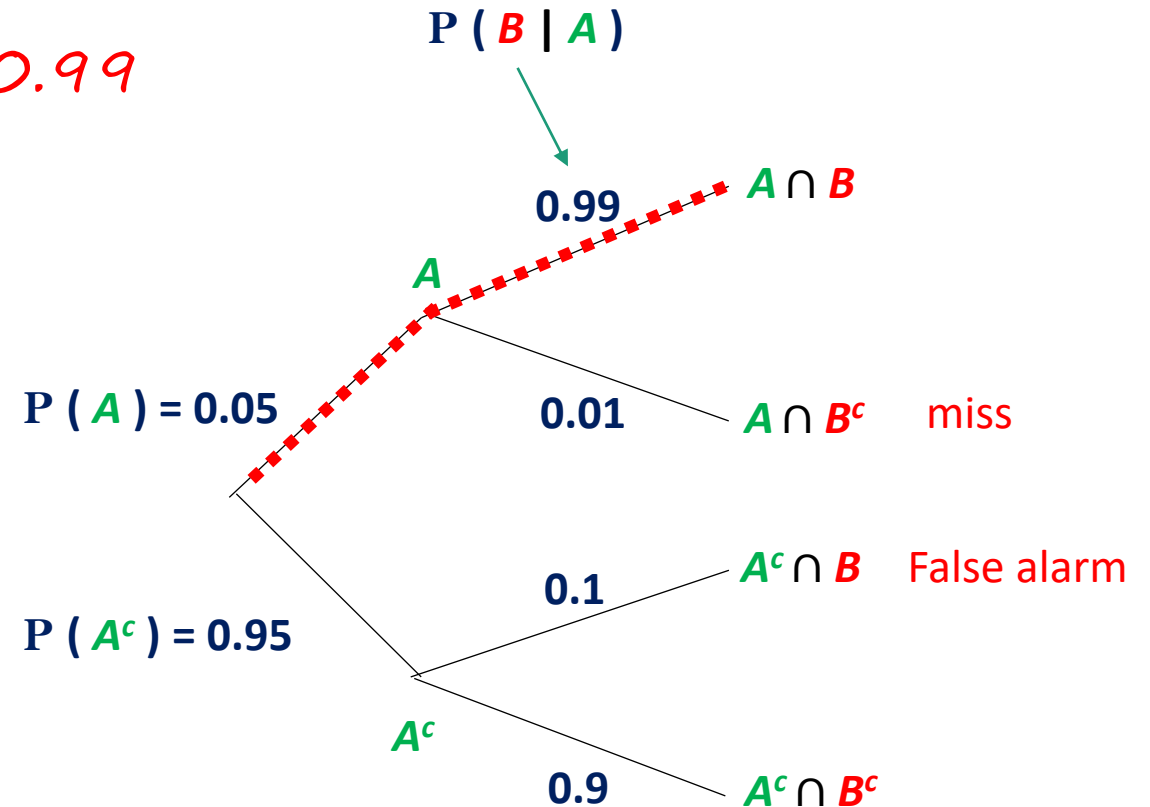
❖ Problem - Find The probability an aircraft is flying given radar registers something on radar screen – $P(A|B)$?

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\rightarrow \bullet P(A \cap B) = P(A) * P(B|A) = 0.05 * 0.99$$

$$\rightarrow \bullet P(B) = 0.05 * 0.99 + 0.95 * 0.1 = 0.1445$$

$$\rightarrow \bullet P(A|B) = \frac{0.05 * 0.99}{0.1445} = 0.34$$



Multiplication Rule

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} P(A \cap B) &= P(A | B) P(B) \\ &= P(B | A) P(A) \end{aligned}$$

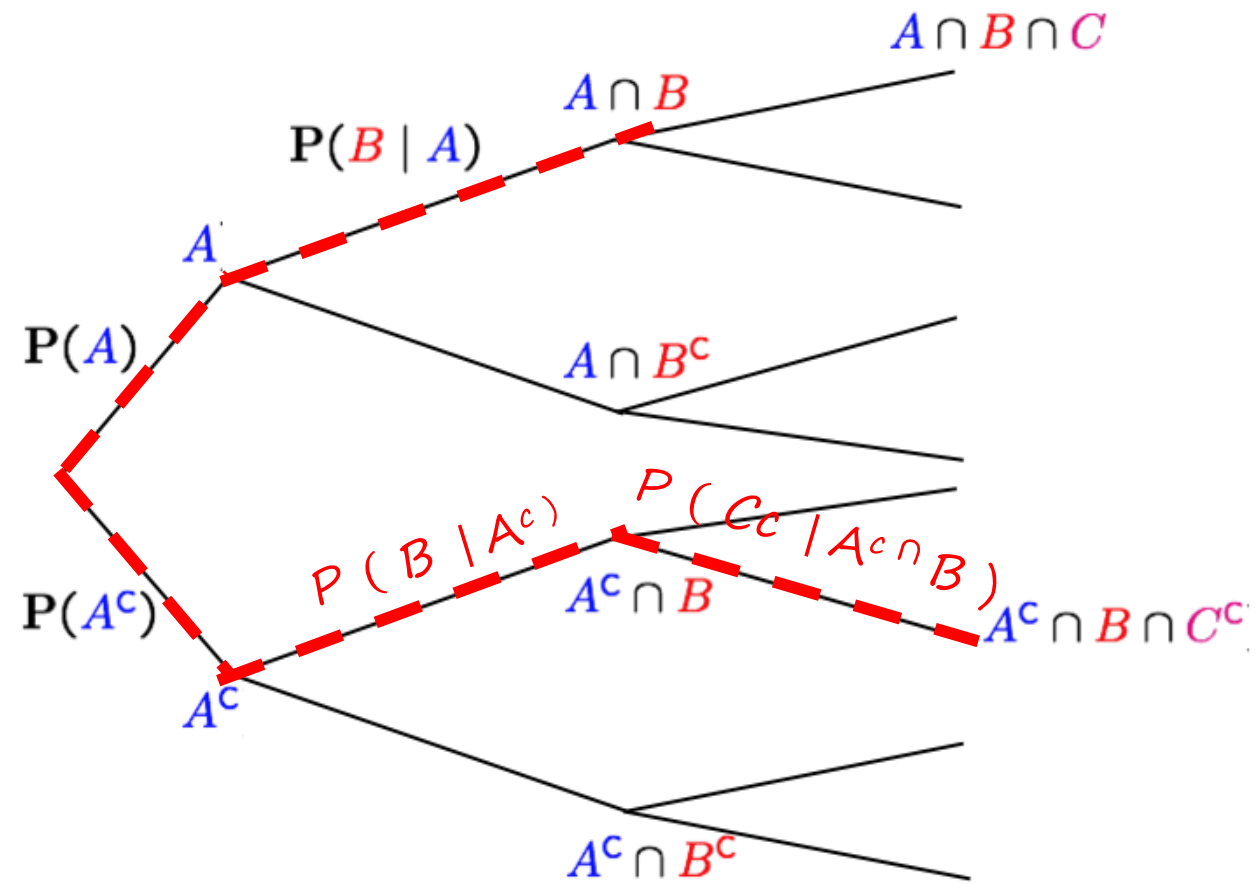
$$P(A^c \cap B \cap C^c) =$$

$$\rightarrow = P(A^c \cap B) P(C^c | A^c \cap B)$$

$$\rightarrow = P(A^c) P(B | A^c) P(C^c | A^c \cap B)$$

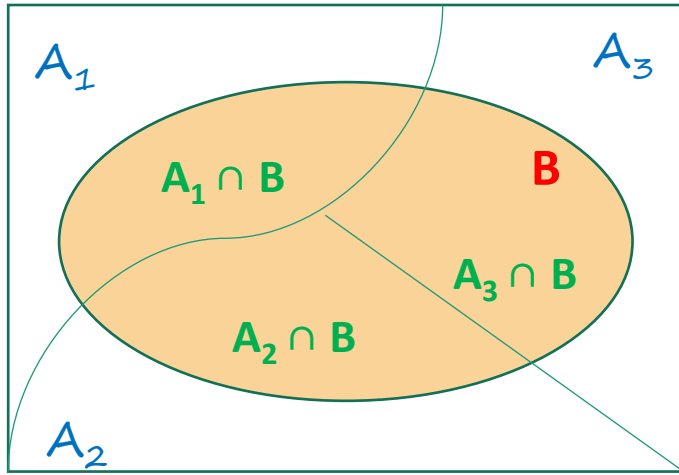
$$\rightarrow P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_4)$$

$$\rightarrow = P(A_1) \prod_{i=2}^n P(A_i | (A_1 \cap A_2 \cap \dots \cap A_{i-1}))$$



Total Probability Theorem

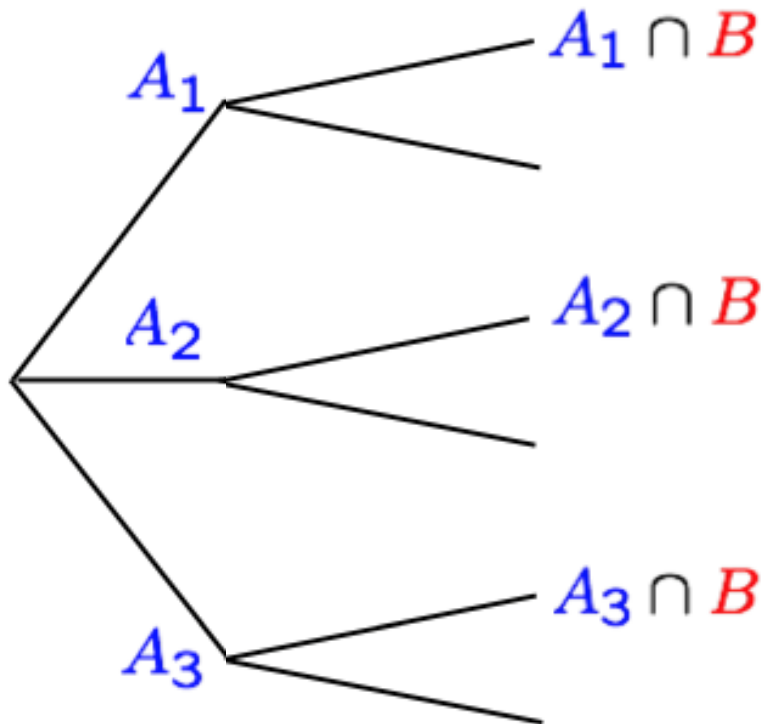
Ω



- Partition of sample space in A_1, A_2, A_3
- Have $P(A_i)$, for all i .
- Have $P(B | A_i)$, for all i .

$$\Rightarrow P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3)$$

$$\Rightarrow = P(A_1)P(B|A_1) + \dots + \dots$$



$$\Rightarrow \sum_j P(A_j) = 1$$

$$P(B) = \sum_j P(A_j) P(B | A_j)$$

weights

weighted average
of $P(B | A_i)$